Retail Customer Analysis

Phase 5 — Customer Lifetime Value Prediction

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Overview

This phase looks forward: it predicts how much each customer is likely to be worth over the next 12 months. Using two established models of buying behaviour, it estimates how often each customer will return, the chance they are still active, and how much they will spend — combined into a single lifetime-value figure. The models were tested against a held-back period of real data before being trusted, to confirm their predictions hold up.

Results at a Glance

£5,244	24x	+8%	4,179	£772
AVG CHAMPION 12-MONTH VALUE	CHAMPION VS HIBERNATING VALUE	HOLDOUT TEST ACCURACY	CUSTOMERS VALUED	MEDIAN CLV

Key Findings

1. **Champions are worth far more than any other group going forward.**

   Each Champion is predicted to be worth about £5,244 over the next year — roughly 24 times a Hibernating customer (£219). The forward-looking model and the behavioural segments agree on who matters most.

2. **The value model was validated before being trusted.**

   Tested against a held-back nine-month period, the model's predicted purchases came within about 8% of what actually happened — close enough to rely on for planning and targeting.

3. **Future value is highly concentrated.**

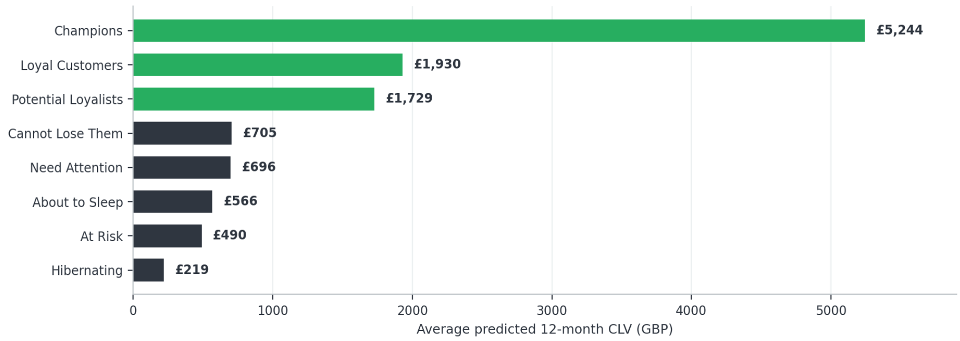
   The median customer is worth around £772 over the year, but a smaller group of frequent, recent buyers is worth many times that. Retention spend is best focused on protecting this high-value core.

4. **Individual predictions are reliable only with enough history.**

   For customers with very few past purchases, the value estimate is unstable — a single large order can inflate it. These cases were flagged, and reliable rankings restricted to customers with a proven repeat history, so decisions rest on trustworthy numbers.

Predicted Value by Segment

Average predicted 12-month value per customer in each segment. Value tracks the behavioural segments from the previous phase — two independent methods agreeing.



What's Next — Phase 6

Phase 6 shifts from who customers are to what they buy together, using market basket analysis to find products frequently purchased in the same order — the basis for cross-selling, bundling and product placement.